

DRONE MATRX

# Product Requirements Document

Autonomous Multi-Drone Swarm Orchestration Platform

|                |                                                          |
|----------------|----------------------------------------------------------|
| Version        | 1.1                                                      |
| Status         | Draft for Mentor Review                                  |
| Date           | June 2026                                                |
| Platform       | Web SaaS · Desktop-first · Field-deployable Edge         |
| Simulation     | Gazebo Garden + PX4 SITL (pre-flight validation)         |
| Market         | AgriTech, Disaster Response, Infrastructure, Defense R&D |
| Classification | Confidential — Internal Use Only                         |

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SECTION 01

# Executive Summary

Drone Matrx is an autonomous multi-drone swarm orchestration platform that lets a single operator manage an entire fleet of drones with minimal input. The operator draws an area of interest on a map — the software does everything else: task allocation via CBBA (Consensus-Based Bundle Algorithm), real-time collision avoidance via ORCA, dynamic replanning, live telemetry streaming, and mission reporting.

Every mission can first be validated in Gazebo Garden + PX4 SITL simulation — with identical CBBA and ORCA logic — before committing to real hardware. Built for AgriTech, disaster response, infrastructure inspection, and defense R&D;, Drone Matrx is plug-and-play with PX4 and ArduPilot flight stacks and designed to operate fully offline via a field-deployable edge server.

## 1 Operator

manages entire drone  
fleet

## CBBA+ORCA

autonomous task and  
deconfliction

## Gazebo SITL

pre-flight simulation fidelity

## Offline-first

decentralised edge architecture

## Plug & Play

PX4 / ArduPilot  
compatible

## SECTION 02

## Problem Statement

Existing drone solutions force operators to manually pilot each unit, rely on constant connectivity, and cannot coordinate fleet-level intelligence.

| #  | Problem                            | Why It Matters                                                                                                             |
|----|------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 01 | <b>Manual single-drone ops</b>     | No fleet-level intelligence in consumer/prosumer GCS tools. Each drone requires a dedicated pilot.                         |
| 02 | <b>No swarm coordination</b>       | Multi-drone missions require manual deconfliction — creating dangerous overlaps and coverage gaps.                         |
| 03 | <b>Connectivity dependency</b>     | Most platforms require continuous cloud connectivity — impossible in disaster zones, border areas, or rural fields.        |
| 04 | <b>No autonomous task division</b> | Operators must manually assign waypoints per drone. CBBA-based automatic task bundling does not exist in accessible tools. |
| 05 | <b>No pre-flight sim fidelity</b>  | Existing simulators don't run the same algorithms as live — bugs surface only in real flight at high cost and risk.        |
| 06 | <b>Poor situational awareness</b>  | Live multi-video walls, detection feeds, and telemetry are siloed across separate dashboards.                              |
| 07 | <b>No offline replay / audit</b>   | Mission data is either not recorded or stored in proprietary formats with no replay or export capability.                  |
| 08 | <b>Fragmented payload control</b>  | Sprayer, thermal, and spotlight payloads require separate apps or physical controllers.                                    |
| 09 | <b>FC pairing complexity</b>       | Binding flight controllers to software is manual, error-prone, and non-standardised across airframes.                      |
| 10 | <b>Compliance gaps</b>             | Geo-fence enforcement, NFZ checks, and audit logs are manual or absent, creating DGCA regulatory risk.                     |

## SECTION 03

## User Personas

Drone Matrx serves four distinct operator profiles simultaneously.

### FA Field Operator / Pilot · AgriTech, Disaster

| Pain Points                                                                                                                                                  | Gains                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Must monitor multiple drones from a tablet in the field, often with poor connectivity. Wants a single-screen view of all drones — battery, position, status. | One-tap AOI draw kicks off autonomous swarm. Validate in Gazebo sim first. Live video wall with DVR. Quick RTH/HOLD per drone or all-fleet. |

### MC Mission Commander · Enterprise / Defense

| Pain Points                                                                                                                                | Gains                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Plans complex multi-phase missions, assigns drone roles, reviews post-mission reports. Needs policy-level control — geofences, NFZs, RBAC. | 6-step planning wizard with CBBA preview, Gazebo simulation review, constraint enforcement, mission replay, and PDF export reports. |

### DA Data Analyst / Inspector · Infrastructure, AgriTech

| Pain Points                                                                                          | Gains                                                                                                                                        |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Needs to extract value from mission data: orthomosaic coverage, detection events, thermal anomalies. | Built-in analytics dashboard, detection feed with accept/reject, clip/snapshot export, structured mission reports with PostGIS spatial data. |

### SA System Admin / Fleet Manager · Any enterprise org

| Pain Points                                                                                    | Gains                                                                                                                          |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Manages drone fleet health, firmware versions, user roles, geofence libraries, plugin updates. | Fleet management with payload calibration status, maintenance logs, RBAC, Gazebo world config, system health, plugin registry. |

## SECTION 04

## Goals & Success Metrics

Eight primary goals with measurable KPIs determining whether v1.1 ships successfully.

| Goal / KPI                         | Target                       | Definition                                                                |
|------------------------------------|------------------------------|---------------------------------------------------------------------------|
| Single-operator fleet management   | 1 operator : $\geq$ 8 drones | One logged-in user can run a full swarm mission without additional pilots |
| Autonomous task allocation latency | < 2 sec                      | CBBA re-convergence after drone loss or AOI update                        |
| Swarm deconfliction response       | < 500 ms                     | ORCA velocity-obstacle resolution per drone on collision course           |
| Live video stream latency          | < 300 ms                     | WebRTC end-to-end from drone camera to operator browser                   |
| Gazebo sim to live parity          | 100% fidelity algorithm      | CBBA + ORCA behave identically in Gazebo and real flight                  |
| Offline mission execution          | 100%                         | Full swarm mission runs without any cloud connectivity via edge server    |
| Mission report generation          | < 90 sec                     | PDF + JSON report with coverage %, detections, battery usage              |
| Concurrent drone support (MVP)     | 1 to 16                      | Normal to max swarm size on single edge server                            |
| Dashboard load time                | < 2 sec                      | On standard broadband for Operator UI                                     |

SECTION 05

# Feature Specifications — Detailed

All features are broken into three priority tiers. Each entry includes the UI surface it lives in, data it reads/writes, edge cases, acceptance criteria, and data model hints to aid mockup and implementation.

|             |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0 - LAUNCH | <b>AOI Draw &amp; CBBA Task Allocation</b> | <p>The foundational operator gesture: draw a polygon on the map and press Start. CBBA computes optimal task bundles per drone in &lt; 2 s.</p> <ul style="list-style-type: none"><li>• UI Surface: Mission Control map panel — Leaflet/Mapbox draw tools (polygon, rectangle, freehand, KML/GeoJSON import).</li><li>• CBBA runs server-side on edge orchestrator; result sent back as per-drone waypoint list via WebSocket.</li><li>• Coverage preview: lane lines drawn on map before mission start, with ETA and battery cycle count per drone.</li><li>• Re-allocation triggered automatically when drone joins or drops; operator notified via toast + fleet wall update.</li><li>• Constraints: max altitude, min overlap %, speed, payload type per zone — configured in wizard Step 3.</li><li>• Data model: Mission { id, aoi_geojson, status, created_at }, TaskBundle { drone_id, waypoints[], eta }.</li><li>• Acceptance: 8-drone AOI allocated in &lt; 2 s on Intel NUC i7 edge hardware.</li><li>• Edge case: AOI smaller than one lane width — system warns and suggests single-drone mode.</li><li>• Edge case: More drones available than needed — surplus drones placed on standby with visual indicator.</li><li>• Edge case: AOI crosses a geofence boundary — wizard Step 6 shows conflict overlay and blocks launch until resolved.</li></ul> |
|-------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|             |                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0 - LAUNCH | <p><b>ORCA Collision Avoidance &amp; Global Deconfliction</b></p> | <p>Two-layer safety: ORCA runs on each companion computer for local avoidance; edge orchestrator handles global deconfliction via altitude banding and priority queues.</p> <ul style="list-style-type: none"><li>• ORCA velocity-obstacle computation: runs every 100 ms per drone agent on companion compute.</li><li>• Global layer: assigns altitude bands per drone (e.g. Drone 1: 50 m, Drone 2: 55 m) to prevent head-on conflicts.</li><li>• Priority queue: RTH drone always gets lowest altitude band; active survey drones yield.</li><li>• Collision prediction alert shown in Operator UI <math>\geq 3</math> s before predicted intersection — amber pulsing line on map.</li><li>• No collision guarantee: validated in Gazebo sim with 16-drone scenario before every live deployment.</li><li>• Data: ORCAEvent { drone_a, drone_b, min_dist, resolution_vector, timestamp }.</li><li>• Edge case: all altitude bands occupied — orchestrator holds new drone at safe site until band frees.</li><li>• Edge case: ORCA computation timeout — drone enters HOLD mode, operator alerted immediately.</li><li>• Acceptance: zero collisions in 50 consecutive 8-drone Gazebo sim runs.</li></ul>                                                                                                                                                                                                                                                                                                                                                  |
| P0 - LAUNCH | <p><b>Live Multi-Drone Map &amp; Video Wall</b></p>               | <p>The primary operator screen during an active mission. Map on left (~65% width), fleet wall and video on right (~35%).</p> <ul style="list-style-type: none"><li>• Map: real-time markers per drone (position, heading cone, battery SOC color ring — green/amber/red).</li><li>• Path rendering: planned path (dashed gray), executed path (solid teal), coverage heatmap overlay toggleable.</li><li>• NFZ polygons: red hatched overlay; drones cannot enter. Geofences: blue dashed outlines.</li><li>• Video wall: 1/4/9/N grid; click to pin any feed to fullscreen; DVR scrub bar per feed; PiP mode for single focus.</li><li>• Fleet wall: per-drone card — battery %, link quality bar (4-level RSSI), GNSS fix icon, altitude, ground speed.</li><li>• Quick actions per drone: HOLD · RTH · Land · Go-To-Point · Assign Task — context menu or icon buttons.</li><li>• All-fleet buttons: HOLD ALL · RTH ALL — must confirm dialog to prevent accidental press.</li><li>• Detections feed: live cards with thumbnail, confidence %, GPS coords, Accept/Reject buttons.</li><li>• Battery swap queue: when SOC &lt; 20%, drone auto-queued for swap — amber badge on fleet card.</li><li>• Edge case: video stream lost — feed shows last frame with red 'LINK LOST' overlay.</li><li>• Edge case: GPS fix lost — drone marker grays out, 'GPS LOST' label shown, map auto-zooms to last known.</li><li>• Mockup notes: split panel layout, dark theme, teal accent highlights. Map should fill 65% on <math>\geq 1440</math>px screens.</li></ul> |

## P0 · LAUNCH

**Mission Planning Wizard  
(6-Step)**

Structured wizard guiding the operator from template to launch-ready mission.

- Step 1 — Template: choose from saved templates or start blank. Templates store all params. Card-grid layout.
- Step 2 — AOI: draw polygon or import file. Map shows terrain, existing geofences, NFZs. Layer toggles in sidebar.
- Step 3 — Flight Params: altitude (AGL/MSL toggle), speed (m/s), overlap %, corridor width, heading lock. Slider + numeric input.
- Step 4 — Sensors: assign payload per drone (camera, thermal, multispec, sprayer); calibration status chip (green/amber/red).
- Step 5 — Constraints: geofences active, NFZ enforcement on/off, link-loss policy dropdown, max flight time, weather hold.
- Step 6 — Validate: CBBA allocation preview on map, battery cycle count, estimated coverage %, NFZ conflict check. Buttons: Save Draft · Run in Gazebo Sim · Launch Live.
- Offline tile cache: map tiles pre-downloaded for planned AOI (configurable radius buffer).
- Wizard state auto-saved every step to IndexedDB — operator can close browser and resume.
- Progress stepper at top with step labels and checkmark for completed steps.
- Edge case: AOI outside cached tile area — warning with option to download tiles now (if online).
- Mockup notes: stepper UI top, main content center, param panel right side. Dark card style on dark bg.

## P0 - LAUNCH

**Gazebo Simulation Mode**  
**— Full Spec**

Full pre-flight validation using Gazebo Garden + PX4 SITL. See Section 07 for architecture detail.

- Gazebo world auto-generated from mission AOI + SRTM terrain elevation data.
- PX4 SITL instances spawned per drone — one process per simulated FC. Port range: 14540+N.
- CBBA and ORCA run identically to live — same code path, same libraries, zero divergence.
- Simulated sensors: camera (Gazebo rendered scene), battery drain model (configurable curve), GNSS (with configurable noise).
- Speed multiplier: run simulation at 1x / 2x / 5x / 10x for rapid validation — slider in sim toolbar.
- Failure injection panel: kill drone, degrade link quality, force low battery, add pop-up obstacle, GPS drift.
- Sim report generated on completion: coverage %, battery per drone, any ORCA interventions, task completion per drone.
- Diff view: compare sim report vs previous live mission report for the same AOI — side-by-side table.
- Gazebo world viewer: embedded 3D viewport or external Gazebo client link with camera presets.
- Sim session saved to DB — replayable with identical controls as live replay.
- Edge case: Gazebo fails to launch (resource constraint) — error dialog with system requirements and suggested fix.
- Edge case: PX4 SITL process crashes mid-sim — treated as drone failure, CBBA re-allocates, logged as simulated event.

## P0 - LAUNCH

**Live Mission Mode with FC Binding**

Real flight with physically bound flight controllers. Full FC integration in Section 06.

- FC binding via UIN auto-discovery (UDP broadcast), QR scan, or manual UIN entry.
- Pre-arm checklist displayed per drone: GPS fix, battery %, IMU calibration, link quality, payload status.
- Arm/disarm confirmation dialog with checklist pass/fail visual — blocked if any critical check fails.
- Mission upload: waypoints uploaded to FC via MAVLink mission protocol with progress bar.
- In-flight: all commands (HOLD, RTH, Land, Go-To) sent via MAVSDK to FC in < 50 ms.
- Failsafe: if orchestrator loses heartbeat from any drone > 2 s, link-loss policy executes automatically.
- Concurrent arm: operator can arm all drones sequentially or staggered (configurable delay 0–10 s).
- Edge case: FC refuses arm (safety interlock) — specific MAVLink error shown with resolution hint.
- Edge case: mission upload fails — retry with exponential backoff; operator notified after 3 failures.
- Acceptance: arm-to-takeoff latency < 3 s per drone on stable link.

## P0 - LAUNCH

**Fleet Management — Drones & Payloads**

Central registry of all drones, companion computers, and payloads in the org.

- Drone card grid: SOC ring (green >50%, amber 20–50%, red <20%), link bars (4-level RSSI), GNSS quality dot, health score badge.
- Filters: status (online/offline/active/maintenance), capability tags, payload type, firmware version.
- Drone detail slide-out: UIN, FC model, firmware version, total flight hours, last maintenance date, autopilot log download button.
- Payload calibration panel: per-payload last-calibrated date, run calibration test (sprayer flow test, camera focus, thermal NUC).
- Maintenance log: free-text entries with timestamp, technician name, and category tag (routine/repair/upgrade).
- Bulk actions toolbar: select multiple drones → assign to mission, update firmware, export fleet report.
- Data model: Drone { uin, fc\_model, fw\_version, total\_hours, status, payloads[] }, Payload { type, calibrated\_at, status }.
- Edge case: firmware mismatch — drone card shows amber warning badge with 'Update Required' label.
- Mockup notes: grid/list toggle view, search bar top, filter chips below. Dark cards on dark background.

|             |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0 · LAUNCH | <b>Role-Based Access Control (RBAC)</b>           | <p>Three roles enforced at API middleware level.</p> <ul style="list-style-type: none"><li>• Operator: can run assigned missions, view own fleet, access Gazebo sim for assigned missions.</li><li>• Mission Commander: full mission CRUD, analytics access, can assign Operators to missions, manage geofences.</li><li>• System Admin: org config, user management, plugin registry, geofence library, all audit logs, fleet management.</li><li>• JWT RS256 auth; access token 1 h, refresh token 7 days. Stored in httpOnly cookies.</li><li>• All API routes have middleware RBAC check — frontend route guards are UX convenience only, not security.</li><li>• Audit log: every destructive action (delete mission, remove drone, change role) logged with timestamp, actor, IP.</li><li>• Multi-org: each org is a fully isolated tenant. Super Admin can view cross-org (Phase 2).</li><li>• Data model: User { id, email, role, org_id }, AuditLog { action, actor_id, target, timestamp }.</li></ul>                                                                                  |
| P0 · LAUNCH | <b>Onboard Communication &amp; Fallback Links</b> | <p>Resilient multi-layer communication stack for field operations.</p> <ul style="list-style-type: none"><li>• Primary: Wi-Fi mesh (802.11s or point-to-multipoint AP). Range ~300 m per hop.</li><li>• Fallback 1: LTE (4G SIM on companion compute). Latency ~80 ms typical.</li><li>• Fallback 2: LoRa (SX1276 module). Heartbeat + basic commands only. Range ~5 km.</li><li>• Auto-fallback: link quality monitored every 500 ms; switch triggers if RSSI drops below configurable threshold.</li><li>• Telemetry: NATS pub/sub over active link. Topics: drone/{UIN}/telemetry, drone/{UIN}/state, drone/{UIN}/alert.</li><li>• Video: WebRTC peer-to-peer over Wi-Fi/LTE. Coturn TURN server for NAT traversal.</li><li>• Link-loss policy (configurable per mission): RTH · HOLD · Continue waypoints · Land at nearest safe site.</li><li>• Edge case: all links lost simultaneously — onboard failsafe on FC (configured pre-flight) executes independently of software.</li><li>• Data model: LinkStatus { drone_id, primary_rssi, fallback_rssi, active_link, timestamp }.</li></ul> |

## P1 · PHASE 2

**Replay — Time-Synced Map & Video**

Full post-mission replay for debrief, audit, and training.

- Replay player: Play, Pause, Scrub, 1x/2x/5x speed, Jump to event — transport bar at bottom.
- Map and video feeds time-synced to within 100 ms of each other.
- Mission timeline strip: markers for events (detection, battery swap, link loss, ORCA intervention), clickable notes.
- Drone track playback: animated markers trace executed path; coverage heatmap fills progressively.
- Detection review: replay pauses at each detection event; operator can re-accept/re-reject with annotation.
- Clip export: select time range on timeline → export MP4 (all active feeds PiP) or PNG snapshot.
- Key metrics sidebar: coverage % achieved, total area surveyed, detections count, battery per drone, incidents list.
- Edge case: video recording missing for a drone — placeholder frame shown; map replay continues unaffected.
- Mockup notes: similar layout to live mission but with transport controls replacing quick actions.

## P1 · PHASE 2

**Reporting & Export**

Auto-generated mission reports for clients, compliance, and data pipelines.

- PDF report sections: mission summary, coverage map (orthomosaic thumbnail or heatmap), per-drone stats, detection log with thumbnails, incident timeline, battery log.
- JSON export: full structured telemetry, detection events, CBBA allocation decisions.
- CSV export: per-timestamp telemetry rows for external analytics tools.
- Reports stored in MinIO with org-scoped signed URLs (7-day link expiry, configurable).
- Audit log: every report access (view, download) logged with user and timestamp.
- Scheduled reports: auto-generate at mission completion and email to configured recipients.
- AgriTech variant: crop stress detection summary, NDVI proxy from multispec data, field coverage map.
- Infrastructure variant: anomaly detection log with GPS coordinates, severity rating, recommended action.
- Mockup notes: report preview modal with tabs for each section. Download buttons: PDF / JSON / CSV.

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1 · PHASE 2 | <p><b>Perception Models &amp; Detection Feed</b></p> <p>Pluggable ML inference pipeline for real-time object and anomaly detection.</p> <ul style="list-style-type: none"><li>• Model interface: ONNX Runtime on companion compute (Jetson preferred). CPU fallback on Pi 4.</li><li>• Bundled models: person detection (YOLOv8n), vehicle detection (YOLOv8n), crop-stress classifier (custom).</li><li>• Custom model upload: drag-and-drop ONNX file in Settings → Plugins. Validated on sim before live use.</li><li>• Detection event: thumbnail, confidence score, GPS coordinate, drone UIN, timestamp, model name.</li><li>• Operator actions per detection: Accept (write to PostGIS) · Reject (discard) · Create Task (send drone to inspect).</li><li>• Confidence threshold: per-model, configurable in Settings. Default 0.65.</li><li>• Detection heatmap: accepted detections rendered as point cloud on map after mission.</li><li>• Edge case: inference FPS drops below 2 — warning in fleet wall; model may be too heavy for hardware.</li><li>• Data model: Detection { id, mission_id, drone_id, model, confidence, geom_point, thumbnail_url, status }.</li></ul> |
| P1 · PHASE 2 | <p><b>Geofences, NFZ &amp; Safe Landing Sites</b></p> <p>Spatial constraint system enforced at planning, simulation, and runtime.</p> <ul style="list-style-type: none"><li>• Geofence types: Inclusion (must stay inside), Exclusion (must stay outside), Altitude ceiling.</li><li>• Import formats: KML, GeoJSON, Shapefile (.shp). Export same formats.</li><li>• NFZ enforcement: mission start blocked if any waypoint inside NFZ. Live: immediate RTH triggered.</li><li>• Safe landing sites: registry of GPS-tagged clearings/pads. Used by link-loss and low-battery policies.</li><li>• Crowd/structure standoff: configurable minimum distance (default 30 m) from detected humans or structures.</li><li>• Geofence library: org-level saved set; reusable across missions. CRUD in Settings page.</li><li>• Gazebo sim: geofences imported into Gazebo world as collision-free zones for simulation fidelity.</li><li>• Edge case: drone exits geofence due to wind — alert shown, RTH triggered after 5 s grace period.</li><li>• Data model: Geofence { id, name, type, geom_polygon, alt_ceiling, org_id }.</li></ul>                                                  |

|              |                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P2 · PHASE 3 | <b>AI Dynamic Replanning</b>                       | <p>ML model observes live mission data and proactively rebalances task assignments mid-mission.</p> <ul style="list-style-type: none"><li>• Triggers: drone battery drain faster than predicted, coverage gap detected, new detection cluster, weather change.</li><li>• Replan proposal shown to operator: 'Drone 3 rerouting to cover gap in sector B. ETA change: +4 min.' Accept / Dismiss.</li><li>• Auto-accept mode: Mission Commander can enable for fully autonomous replanning (opt-in per mission).</li><li>• Replan logged to audit trail with reason, old allocation, new allocation, and operator response.</li><li>• Acceptance: replanning proposal generated in &lt; 3 s from trigger event.</li></ul> |
| P2 · PHASE 3 | <b>Advanced Analytics Dashboard</b>                | <p>Fleet-level analytics for enterprise clients and platform improvement.</p> <ul style="list-style-type: none"><li>• Fleet KPI tiles: missions this month, avg coverage %, avg battery efficiency, total flight hours.</li><li>• Coverage efficiency trend: line chart per week/month with drill-down per mission.</li><li>• Detection accuracy: acceptance rate per model, false-positive trend line.</li><li>• Battery cycle health: per-drone SOC degradation over time (scatter plot).</li><li>• Exportable for AgriTech crop reports and infrastructure inspection compliance filings.</li><li>• Mockup notes: grid of KPI cards at top, charts below in 2-col layout. Date range picker top-right.</li></ul>     |
| P2 · PHASE 3 | <b>Public API / SDK + Third-Party Integrations</b> | <p>External integration layer for enterprise and government customers.</p> <ul style="list-style-type: none"><li>• REST API: mission CRUD, fleet status, detection events, report download — OpenAPI 3.0 spec.</li><li>• WebSocket: subscribe to live telemetry stream for any drone or mission.</li><li>• Webhook triggers: mission started, mission completed, detection event, link-loss alert.</li><li>• SDK: Python + Node.js packages for custom payload drivers and perception models.</li><li>• API key management: per-user, scoped to read/write, with rate limiting (1000 req/min default).</li></ul>                                                                                                        |

SECTION 06

# Flight Controller Integration Strategy

How the software identifies, binds, and communicates with real flight controllers in the field — a core product decision for a hardware-software platform.

## Recommended Approach: MAVLink Heartbeat + UIN Registry

Each physical flight controller (PX4 or ArduPilot) broadcasts a MAVLink HEARTBEAT message every second over the active link. The onboard agent (MAVSDK process on a companion computer) picks up this heartbeat and registers the FC with the edge orchestrator using a Unique Identifier Number (UIN).

### UIN Structure

| UIN Component | Format       | Example       | Notes                                       |
|---------------|--------------|---------------|---------------------------------------------|
| Org Prefix    | 3-char alpha | DRN           | Assigned by Drone Matrx on org creation     |
| Fleet ID      | 2-digit      | 04            | Fleet number within org                     |
| Drone Serial  | 6-digit      | 210935        | Airframe serial or operator-assigned        |
| Full UIN      | —            | DRN-04-210935 | Stickered on airframe + stored in companion |

### Three FC Binding Methods

#### Method A — Auto-Discovery (recommended for field)

Companion compute broadcasts UIN on UDP port 14550 over local Wi-Fi mesh. Operator UI shows 'Available Drones' panel automatically populated. One-click bind. Works fully offline.

- Companion runs avahi/mDNS + UDP beacon every 2 s.
- Operator UI listens on same subnet. No manual IP entry required.
- Drone removed from available list after 10 s without beacon (assumed powered off).

#### Method B — QR Code Scan

Each companion computer has a QR sticker encoding the UIN + current IP endpoint. Operator scans with phone using Drone Matrx mobile helper app → pushed to desktop UI.

- QR encodes: dm://bind?uin=DRN-04-210935&ip;=192.168.1.42&port;=14550
- Mobile helper app passes bind request to operator's browser via WebSocket.
- QR refreshes IP on each boot — always current.

#### Method C — Manual UIN Entry

Operator types or pastes the UIN in the Fleet screen. System resolves endpoint from the UIN registry or prompts for IP. Fallback for edge cases.

- System checks UIN against registry for last-known IP as starting point.
- Operator can override IP manually (useful for static IP fleets).
- Connection verified with one HEARTBEAT exchange before confirming bind.

### FC Health Monitoring Post-Bind

- HEARTBEAT timeout > 2 s triggers link-loss policy immediately.
- FC firmware version checked against approved whitelist on bind — mismatch shows amber warning.
- IMU calibration status, GPS fix quality, battery cell voltages shown in Fleet detail pane.

- Pre-arm checklist: GPS fix (3D lock required), battery > 30%, no active failsafes, payload calibrated.
- All MAVLink messages logged to TimescaleDB for post-mission replay and audit trail.
- FC disconnect during mission: orchestrator waits 2 s → executes link-loss policy → logs event.

SECTION 07

# Mission Modes — Gazebo Simulation vs Live Flight

Every mission in Drone Matrx operates in one of two modes, explicitly selected by the operator before launch. The core design principle is algorithmic parity: the same CBBA allocation code, the same ORCA deconfliction engine, and the same mission planning wizard produce missions that execute identically in simulation and in the real world. There is no separate sim codebase.

*ALGORITHM PARITY GUARANTEE: CBBA and ORCA modules are imported identically by both the Gazebo simulation bridge and the live edge orchestrator. The only difference is the telemetry source — simulated sensor data from Gazebo vs real MAVLink from hardware. This guarantees a mission validated in Gazebo will behave identically in real flight.*

## MODE A — Gazebo Garden + PX4 SITL Simulation

The simulation environment is powered by Gazebo Garden (or Harmonic) with PX4 SITL running one PX4 process per simulated drone. The edge orchestrator connects to each SITL instance via MAVSDK exactly as it would connect to real hardware — the orchestrator is agnostic to whether it is talking to a physical FC or a simulated one.

### Gazebo World Generation Pipeline

- AOI polygon from mission wizard is projected into Gazebo world coordinates (ENU frame).
- SRTM terrain elevation data (30 m resolution) is downloaded or loaded from cache for the AOI bounding box.
- Terrain converted to Gazebo heightmap (.png) using GDAL — placed as ground plane in the world SDF.
- Obstacles (buildings, trees, power lines) can be optionally loaded from OpenStreetMap or manually placed.
- Drone spawn points computed from CBBA allocation results — each drone placed at its assigned start position.
- World SDF file generated programmatically and saved to sim session directory for reproducibility.
- Weather conditions: wind speed/direction configurable per simulation session (affects PX4 SITL physics model).

### PX4 SITL Instance Management

- Each simulated drone gets its own PX4 SITL process, spawned by the sim orchestrator.
- Port allocation: MAVLink UDP port 14540 + drone\_index (e.g., Drone 0: 14540, Drone 1: 14541, etc.).
- PX4 SITL communicates with Gazebo via the px4\_ros\_com bridge or direct Gazebo transport plugin.
- MAVSDK connects to each SITL instance identically to how it connects to real FCs — same API calls.
- Process lifecycle: spawned at sim start, killed at sim end. Crash handled as simulated drone failure.
- Max concurrent SITL instances: 16 on Intel NUC i7 (64 GB RAM recommended), 8 on i5 (32 GB RAM).
- Resource monitoring: CPU and RAM usage per SITL process displayed in sim status bar.

### Simulated Sensor Models

| Sensor       | Gazebo Plugin             |              | Fidelity                          |          | Configurable Parameters                   |  |
|--------------|---------------------------|--------------|-----------------------------------|----------|-------------------------------------------|--|
| Camera (RGB) | gz-sensors camera         |              | Rendered                          | scene at | Resolution, FOV, FPS (5–30)               |  |
|              |                           |              | configurable FPS                  |          |                                           |  |
| Thermal      | gz-sensors thermal_camera |              | Heat map overlay on scene objects |          | Sensitivity, palette (ironbow/rainbow)    |  |
| Battery      | Custom model              | linear drain | SOC decreases                     | by       | Capacity (mAh), drain rate, voltage curve |  |
|              |                           |              | configurable rate                 |          |                                           |  |

|           |                         |                                |                                                        |
|-----------|-------------------------|--------------------------------|--------------------------------------------------------|
| GNSS      | gz-sensors navsat       | GPS coordinates with           | Fix type (2D/3D), HDOP, noise sigma configurable noise |
| IMU       | gz-sensors imu          | Accelerometer + gyroscope data | Noise parameters, bias drift                           |
| Barometer | gz-sensors air_pressure | Altitude estimation from       | Reference pressure, noise                              |

Failure Injection Framework

A critical advantage of Gazebo simulation is the ability to inject controlled failures to validate the system's resilience before committing to real flight. The failure injection panel is accessible from the simulation toolbar during any active sim session.

| Failure Type            | Trigger Method              | Expected System Response                                                      | Validation Criteria                               |
|-------------------------|-----------------------------|-------------------------------------------------------------------------------|---------------------------------------------------|
| Drone kill (motor stop) | Click drone → Kill          | CBBA re-allocates tasks to remaining drones. Killed drone removed from fleet. | Re-allocation in < 2 s. Coverage recalculated.    |
| Link degradation        | Slider: 100% → 10% quality  | Auto-fallback to next link tier. Telemetry rate reduced.                      | Fallback triggers within 1 s of threshold breach. |
| Low battery force       | Set SOC to target %         | Drone auto-queued for RTH at < 20%. Battery swap queue updates.               | RTH triggered within 500 ms of SOC threshold.     |
| GPS drift / loss        | Toggle GPS noise / kill fix | Drone enters HOLD. 'GPS LOST' shown on map. Pos estimation from IMU.          | HOLD mode within 1 s. Operator alerted.           |
| Obstacle spawn          | Click on map to place       | ORCA detects obstacle, drones reroute around it.                              | No collision. Reroute computed in < 500 ms.       |
| Wind gust               | Set speed (m/s) + direction | PX4 compensates. If beyond limits, RTH triggered.                             | Drone maintains course or safely RTHs.            |
| Companion crash         | Kill companion process      | Drone FC onboard failsafe activates. Orchestrator logs loss.                  | FC-level failsafe (RTH/Land) within 2 s.          |

Simulation Speed Control & Performance

- Speed multiplier slider: 1x (real-time), 2x, 5x, 10x. Affects Gazebo physics step rate.
- At 10x speed, an 8-drone 20-minute survey mission validates in ~2 minutes.
- Performance guardrail: if simulation frame rate drops below 10 FPS, speed auto-reduces to maintain fidelity.
- Sim session can be paused at any point — all drone states frozen, inspection possible.
- Sim session can be saved and resumed later (serializes world state + all SITL states).

Simulation Reports & Comparison

- Sim report generated automatically at sim completion: coverage %, battery per drone, ORCA interventions, task completion matrix.
- Report format mirrors live mission report — identical sections and metrics for direct comparison.
- Diff view: side-by-side comparison of sim report vs previous live report for the same AOI.
- Highlighted deltas: coverage difference, battery usage difference, any new ORCA events.

- Sim reports stored in same DB as live reports — tagged with mode='simulation' for filtering.

MODE B — Live Flight with Real Hardware

Live mode connects to physical flight controllers via MAVSDK over the multi-layer communication stack (Wi-Fi mesh, LTE, LoRa). The orchestrator treats live FCs identically to SITL FCs — the only difference is the transport layer and the requirement for pre-flight safety checks.

Pre-Flight Checklist (Live Mode Only)

| Check Item          | Threshold                 | Blocking? | Resolution                                    |
|---------------------|---------------------------|-----------|-----------------------------------------------|
| GPS fix quality     | 3D fix, HDOP < 2.0        | Yes       | Wait for fix or reposition drone to clear sky |
| Battery SOC         | > 30%                     | Yes       | Replace battery before arming                 |
| IMU calibration     | Valid (no errors)         | Yes       | Run IMU calibration in FC settings            |
| Compass calibration | Valid, no interference    | Yes       | Recalibrate away from metal structures        |
| Link quality (RSSI) | > -70 dBm primary         | Warning   | Move closer or check antenna alignment        |
| FC firmware version | In approved whitelist     | Warning   | Update firmware via Fleet Management          |
| Payload status      | Calibrated recently       | Warning   | Run payload calibration test                  |
| Geofence loaded     | Active geofences uploaded | Yes       | Upload geofences from mission config          |
| Weather conditions  | Wind < max threshold      | Warning   | Wait for conditions to improve                |

Live Mission Lifecycle

- 1. Mission created via wizard (Steps 1–6) and validated in Gazebo sim (recommended).
- 2. Operator selects 'Launch Live' → FC binding screen: available drones listed from auto-discovery.
- 3. Drones bound to mission slots. Pre-flight checklist displayed per drone.
- 4. All checks green → 'ARM ALL' button enabled. Operator confirms arm dialog.
- 5. Waypoints uploaded to each FC via MAVLink mission protocol (progress bar per drone).
- 6. Takeoff command sent. Drones take off sequentially (configurable stagger delay).
- 7. CBBA allocation active. ORCA monitoring active. Operator sees live map + fleet wall.
- 8. Mission auto-completes when all tasks done. Report generated. Drones RTH.
- 9. Post-mission: replay available immediately. Reports downloadable.

Mode Comparison Matrix

| Aspect               | Gazebo + PX4 SITL           | Live Flight                          |
|----------------------|-----------------------------|--------------------------------------|
| Flight Controller    | PX4 SITL process (software) | Physical PX4/ArduPilot FC (hardware) |
| Pre-flight checklist | Skipped (not needed)        | Mandatory — blocks arm if failed     |
| Sensor data source   | Gazebo simulated sensors    | Real onboard sensors via MAVLink     |
| CBBA / ORCA code     | Identical (shared library)  | Identical (shared library)           |

|                   |                                  |                                  |
|-------------------|----------------------------------|----------------------------------|
| Failure injection | Full panel available             | Not available (safety)           |
| Speed control     | 1x / 2x / 5x / 10x               | 1x only (real-time)              |
| Risk              | Zero — software only             | Real — requires safety protocols |
| Communication     | localhost / loopback             | Wi-Fi mesh / LTE / LoRa          |
| Video             | Gazebo rendered camera           | Real camera feed via WebRTC      |
| Report            | Sim report (tagged 'simulation') | Live report (tagged 'live')      |
| Cost              | Compute resources only           | Battery, wear, operational cost  |

SECTION 09

# System Architecture

Decentralised, offline-first architecture with edge server as the operational brain.

## Architecture Layers

| Layer              | Components                                                                          | Responsibility                                                                                       |
|--------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Operator (Browser) | UI React + Vite + Tailwind CSS, Leaflet/Mapbox, WebRTC client, Zustand state        | Mission planning, live monitoring, replay, analytics. Connects to edge via WebSocket/REST.           |
| Edge Orchestrator  | Node.js + Fastify, CBBA engine, ORCA engine, NATS client, MAVSDK bindings           | Task allocation, deconfliction, FC communication, telemetry aggregation, mission lifecycle.          |
| Gazebo Bridge      | Sim Python + gz-sim bindings, PX4 SITL process manager                              | Spawns Gazebo worlds and SITL instances. Bridges sim data to orchestrator via same MAVSDK interface. |
| Companion Compute  | MAVSDK agent, ONNX Runtime, link manager (Wi-Fi/LTE/LoRa), camera streamer          | Onboard per-drone. Runs ORCA local layer, perception inference, video stream, telemetry relay.       |
| Data Layer         | PostgreSQL + PostGIS, TimescaleDB (telemetry), MinIO (media/reports), Redis (cache) | Persistent storage. PostGIS for spatial queries. TimescaleDB for time-series. MinIO for blobs.       |
| Message Broker     | NATS JetStream                                                                      | Pub/sub for real-time telemetry, commands, alerts. Durable streams for replay.                       |

## Communication Flow Diagram (Text)

- Operator UI ↔ [WebSocket/REST] ↔ Edge Orchestrator ↔ [NATS] ↔ Companion Compute ↔ [MAVLink] ↔ Flight Controller
- Operator UI ↔ [WebSocket] ↔ Edge Orchestrator ↔ [MAVSDK] ↔ Gazebo SITL (sim mode only)
- Companion Compute → [WebRTC] → Operator UI (video stream, direct peer-to-peer)
- Edge Orchestrator → [NATS] → TimescaleDB (telemetry persistence)
- Edge Orchestrator → MinIO (reports, recordings, detection thumbnails)

## Offline-First Design Principles

- Edge server is the complete operational brain — no cloud dependency for any mission-critical function.
- Map tiles pre-cached for planned AOI (Leaflet offline tile layer with configurable radius).
- All mission data persisted locally (PostgreSQL + MinIO on edge server).
- Cloud sync is optional and deferred — used only for org-wide analytics, firmware updates, and model downloads.
- Service worker on Operator UI caches static assets for browser-level offline resilience.

SECTION 10

# Implementation Roadmap & Detailed Sprints

Four-phase roadmap broken into 2-week sprints. Each sprint lists deliverables, acceptance criteria, and dependencies. Total estimated timeline: 24 weeks (6 months).

## Phase Overview

| Phase                    | Duration   |                 | Focus                                        |  | Key Deliverables                                                                         |
|--------------------------|------------|-----------------|----------------------------------------------|--|------------------------------------------------------------------------------------------|
| Phase 1: Foundation      | 1: Sprints | 1–4 (8 weeks)   | (8 Core infrastructure, auth, basic UI shell |  | Project setup, auth system, map integration, fleet management skeleton, design system    |
| Phase 2: Core Algorithms | 2: Sprints | 5–8 (8 weeks)   | (8 CBBA, ORCA, Gazebo sim, mission wizard    |  | CBBA engine, ORCA engine, Gazebo integration, mission planning wizard, live mission mode |
| Phase 3: Operations      | 3: Sprints | 9–10 (4 weeks)  | (4 Live flight, replay, reporting            |  | FC binding, live telemetry, replay system, report generation, detection feed             |
| Phase 4: Polish & Scale  | 4: Sprints | 11–12 (4 weeks) | (4 Analytics, API, testing, deployment       |  | Analytics dashboard, public API, E2E testing, edge deployment, documentation             |

## Sprint 1 (Weeks 1–2) — Project Setup & Design System

| Deliverable                        | Details                                                                                                                                                     |  | Acceptance Criteria                                                    |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------|
| Vite + React + TypeScript scaffold | Initialize project with Vite, React 19, TypeScript, Tailwind CSS 4, Zustand, Framer Motion. ESLint + Prettier configured.                                   |  | npm run dev starts without errors. Build produces dist/                |
| Design system implementation       | CSS custom properties for all brand colors, typography scale (Inter font), spacing tokens, component primitives (Button, Card, Input, Badge, Modal, Toast). |  | All primitives render correctly. Dark theme applied globally.          |
| Layout shell with routing          | App shell: sidebar nav, top bar, content area. React Router with routes: /, /mission, /fleet, /sim, /analytics, /settings, /login.                          |  | All routes render placeholder pages. Sidebar highlights active route.  |
| Authentication (JWT) flow          | Login page, JWT RS256 token handling (access + refresh), protected route wrapper, mock auth API.                                                            |  | Login, logout, token refresh work. Protected routes redirect to login. |
| Landing page                       | Hero section with animated drone swarm particles, feature cards, stats bar, architecture diagram section. Responsive.                                       |  | Page loads in < 2s. Animations smooth at 60 FPS. Responsive on mobile. |

## Sprint 2 (Weeks 3–4) — Map Integration & Fleet Management

| Deliverable | Details | Acceptance Criteria |
|-------------|---------|---------------------|
|-------------|---------|---------------------|

|                                        |                                                                                                                                                               |                                                                                         |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Map component (Leaflet/Mapbox)         | Dark-themed map with custom tiles. Draw tools (polygon, rectangle, freehand). Layer toggle (heatmap, paths, geofences, NFZ). Drone markers with heading cone. | Draw polygon on map, get GeoJSON output. Layers toggle independently.                   |
| Fleet Management page                  | Drone card grid/list view. Search + filter chips. Drone detail slide-out panel. Payload tab. Maintenance log tab.                                             | CRUD operations on drone records. Filters work correctly. Responsive grid.              |
| Drone state management (Zustand)       | SwarmStore: drones[], missions[], activeView, drone CRUD actions, simulated telemetry generator for dev.                                                      | Store updates propagate to UI in < 16 ms. Simulated telemetry generates realistic data. |
| Database schema (PostgreSQL + PostGIS) | Tables: users, orgs, drones, payloads, missions, task_bundles, geofences, audit_logs. PostGIS extension for geom columns.                                     | Migrations run cleanly. Seed data loads. Spatial queries return correct results.        |
| API layer (Fastify)                    | REST endpoints: /auth/*, /drones/*, /missions/* with RBAC middleware. WebSocket setup for telemetry.                                                          | All endpoints return correct responses. RBAC blocks unauthorized access.                |

Sprint 3 (Weeks 5–6) — CBBA Engine & Mission Wizard

| Deliverable                       | Details                                                                                                                                                  | Acceptance Criteria                                                                   |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| CBBA task allocation engine       | Consensus-Based Bundle Algorithm implementation. Input: AOI polygon + drone list + constraints. Output: per-drone waypoint bundles. Runs on edge server. | 8-drone AOI allocated in < 2 s. Re-allocation on drone add/remove in < 2 s.           |
| Mission Planning Wizard (6 steps) | Full wizard: Template, AOI draw, Flight params, Sensors, Constraints, Validate. Stepper UI. Auto-save per step. CBBA preview on Step 6.                  | Complete wizard flow end-to-end. All params persisted. CBBA preview renders on map.   |
| CBBA visualization on map         | Colored lane lines per drone on map. ETA labels. Battery cycle indicators. Coverage % display.                                                           | Lanes render with distinct colors per drone. ETA updates on param change.             |
| Mission state machine             | Mission lifecycle: DRAFT → PLANNING → VALIDATING → ARMED → ACTIVE → COMPLETED / ABORTED. State transitions enforced.                                     | Invalid transitions rejected. State changes trigger UI updates and audit log entries. |
| Geofence management               | Geofence library CRUD. Import KML/GeoJSON. Map visualization (inclusion = blue, exclusion = red). NFZ enforcement check.                                 | Import/export works. NFZ conflict detected and blocks launch in wizard Step 6.        |

Sprint 4 (Weeks 7–8) — ORCA Engine & Communication Stack

| Deliverable | Details | Acceptance Criteria |
|-------------|---------|---------------------|
|-------------|---------|---------------------|

|                                     |                                                                                                                                                           |                                                                                            |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| ORCA collision avoidance engine     | Optimal Reciprocal Collision Avoidance implementation. Runs per-drone at 10 Hz. Altitude banding for global deconfliction. Priority queue for RTH drones. | Zero collisions in 50 consecutive 8-drone test runs. Resolution in < 500 ms.               |
| ORCA visualization                  | Collision prediction lines (amber pulse) on map when drones approach. ORCA event markers in timeline. Avoidance vectors shown as arrows.                  | Prediction alerts appear 3+ s before potential intersection. Vectors render correctly.     |
| NATS messaging integration          | NATS JetStream setup. Topics: drone/{UIN}/telemetry, /state, /alert, /command. Consumer groups for orchestrator and UI.                                   | Messages delivered in < 50 ms. Durable replay works for last 24 h of messages.             |
| Communication fallback manager      | Link quality monitoring (RSSI polling at 2 Hz). Auto-fallback: Wi-Fi → LTE → LoRa. Link status displayed per drone in fleet wall.                         | Fallback triggers within 1 s of threshold breach. UI shows active link type per drone.     |
| Telemetry persistence (TimescaleDB) | Hypertable for drone telemetry: timestamp, drone_id, lat, lon, alt, speed, heading, soc, rssi. 1 Hz write rate per drone.                                 | 16-drone stream writes without backpressure. Queries for 1-hour window return in < 500 ms. |

## Sprint 5 (Weeks 9–10) — Gazebo Simulation Integration

| Deliverable                      | Details                                                                                                                             | Acceptance Criteria                                                               |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Gazebo world generation pipeline | Python service: takes AOI polygon + SRTM data → generates Gazebo world SDF with terrain heightmap. Drone spawn positions from CBBA. | World generates in < 30 s for 1 km² AOI. Terrain visually matches real elevation. |
| PX4 SITL process manager         | Spawns/kills PX4 SITL instances per drone. Port allocation. Health monitoring. Crash detection and handling.                        | 16 SITL instances launch on NUC i7 (64 GB). Crash handled as drone failure event. |
| Sim bridge (MAVSDK → Gazebo)     | Orchestrator connects to SITL via MAVSDK. Same code path as live mode. Telemetry routed through NATS identically.                   | Orchestrator cannot distinguish SITL from real FC. All commands work identically. |
| Failure injection panel          | UI panel: kill drone, link degrade slider, force battery, GPS drift toggle, obstacle spawn (click-on-map), wind gust settings.      | Each failure type triggers expected system response (see Section 07 table).       |
| Simulation speed control         | Gazebo physics step rate control: 1x/2x/5x/10x. Auto-downscale on low FPS. Pause/resume functionality.                              | 10x speed produces identical results to 1x. FPS guardrail activates below 10 FPS. |
| Sim report generation            | Auto-generate report on sim completion. Same format as live report. Diff view vs previous live report.                              | Report generated in < 30 s. Diff highlights coverage and battery deltas.          |

## Sprint 6 (Weeks 11–12) — Mission Control Live & Video

| Deliverable                    | Details                                                                                                                     | Acceptance Criteria                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Mission dashboard (full)       | Control 3-panel layout: sidebar (nav + fleet), center (map), right (video + detections). All live data from NATS/WebSocket. | Dashboard loads in < 2 s. Telemetry updates at 1 Hz. Smooth at 60 FPS.                      |
| FC binding flow                | Auto-discovery panel, QR scan integration, manual UIN entry. Pre-arm checklist per drone. Arm/disarm confirmation.          | Auto-discovery shows drones within 5 s. Checklist blocks arm on failure.                    |
| Video wall (WebRTC)            | Multi-feed grid (1/4/9). Click to fullscreen. DVR scrub bar per feed. PiP mode. Quality indicator. Link-loss overlay.       | < 300 ms latency on Wi-Fi. Graceful degradation on LTE. Link-loss overlay appears in < 2 s. |
| Quick actions & fleet commands | Per-drone: HOLD/RTH/Land/GoTo/AssignTask.<br>All-fleet: HOLD ALL/RTH ALL with confirm. Emergency LAND ALL.                  | Commands reach FC in < 50 ms. Confirm dialog prevents accidental ALL actions.               |
| Detection feed (live)          | Scrollable cards: thumbnail, confidence, GPS, Accept/Reject buttons. Accepted → PostGIS. Heatmap overlay option.            | Detections appear in < 1 s of inference. Accept/Reject updates DB immediately.              |

## Sprint 7 (Weeks 13–14) — Replay & Reporting

| Deliverable              | Details                                                                                                                         | Acceptance Criteria                                                        |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Replay system            | Transport controls (play/pause/scrub/speed). Time-synced map + video. Timeline strip with event markers. Detection review mode. | Map and video synced within 100 ms. Events clickable. Speed change smooth. |
| Mission report generator | PDF report: summary, coverage map, per-drone stats, detection log, incident timeline, battery log. JSON and CSV exports.        | PDF generated in < 90 s. All sections populated. JSON schema validated.    |
| Report storage & access  | MinIO storage. Org-scoped signed URLs (7-day expiry). Audit log for all report access.                                          | Signed URLs work correctly. Expired links return 403. Access logged.       |
| Clip & snapshot export   | Select time range on timeline. Export MP4 (PiP all feeds) or PNG snapshot. Progress bar during export.                          | MP4 export for 5-min clip in < 2 min. PNG snapshot instant.                |

## Sprint 8 (Weeks 15–16) — Perception & Geofence Enforcement

| Deliverable      | Details                                                                                                                           | Acceptance Criteria                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| ONNX integration | Runtime Inference pipeline on companion compute. Bundled models: YOLOv8n (person, vehicle), crop-stress classifier. CPU fallback. | Inference at $\geq 5$ FPS on Jetson Nano. $\geq 2$ FPS on Pi 4. Models load in < 5 s. |

|                              |                                                                                                                |                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Custom model upload          | Drag-and-drop ONNX file in Settings → Plugins. Validation run on sim before live. Confidence threshold config. | Upload, validate on sim, deploy to companion. Threshold adjustable per model.    |
| Runtime geofence enforcement | Live geofence check at 1 Hz per drone. Inclusion/Exclusion/Altitude ceiling. RTH on breach after 5 s grace.    | Breach detected within 1 s. RTH triggered after grace period. Alert shown in UI. |
| Safe landing site registry   | GPS-tagged sites. Used by link-loss and low-battery policies. Map markers. CRUD in Settings.                   | Nearest safe site calculated in < 100 ms. Used correctly by failsafe logic.      |

## Sprint 9 (Weeks 17–18) — Analytics & Settings

| Deliverable                 | Details                                                                                                                     | Acceptance Criteria                                                        |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Analytics dashboard         | KPI cards, coverage trend chart, detection accuracy chart, battery health chart, mission timeline Gantt. Date range picker. | Charts render with real data. Drill-down works. CSV export per chart.      |
| Settings pages (all tabs)   | Organization, Users & Roles, Geofence Library, Plugin Registry, System Health, Audit Logs. Full CRUD.                       | All settings persist. RBAC enforced. Audit logs filterable and exportable. |
| RBAC enforcement (complete) | All API routes validated. Frontend route guards. Role-based menu visibility. Audit logging for all destructive actions.     | Operator cannot access Admin pages. All destructive actions logged.        |
| Notification system         | Toast notifications for: drone alerts, mission events, system warnings. Notification center with history.                   | Toasts appear within 500 ms. History persisted. Read/unread state tracked. |

## Sprint 10 (Weeks 19–20) — Integration Testing & Hardening

| Deliverable              | Details                                                                                                                            | Acceptance Criteria                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| End-to-end test suite    | Playwright E2E tests: login flow, mission wizard, sim run, replay, report download. CI/CD pipeline.                                | All E2E tests pass. Pipeline runs on every PR. < 10 min total test time.     |
| Gazebo sim stress test   | 16-drone sim with all failure injections. CBBA re-allocation under load. ORCA validation. Memory leak testing.                     | Zero crashes in 10 consecutive 16-drone runs. No memory leaks over 1 h.      |
| Performance optimization | Lighthouse audit. Bundle size optimization (code splitting, lazy loading). Map rendering optimization. WebSocket message batching. | Lighthouse score > 90. Bundle < 500 KB (gzipped). Map smooth with 16 drones. |
| Offline mode testing     | Full mission lifecycle without internet. Map tile cache. Data sync on reconnect. Service worker validation.                        | Complete mission offline. Data syncs correctly on reconnect.                 |

## Sprint 11 (Weeks 21–22) — API & Documentation

| Deliverable             | Details                                                                                                  | Acceptance Criteria                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Public REST API         | OpenAPI 3.0 spec. Endpoints: missions, fleet, detections, reports. API key management. Rate limiting.    | Swagger UI accessible. All endpoints documented. Rate limiting enforced. |
| WebSocket API           | Subscribe to live telemetry. Mission events stream. Connection management. Heartbeat.                    | Client can subscribe and receive telemetry < 100 ms after event.         |
| Developer documentation | Architecture guide, API reference, deployment guide, contribution guide. Hosted on Docusaurus.           | All docs reviewed. Getting-started runs successfully on clean machine.   |
| User documentation      | Operator handbook: mission planning, sim usage, live flight, replay, reports. Screenshots for each flow. | Handbook covers all features. Screenshots match current UI.              |

## Sprint 12 (Weeks 23–24) — Deployment & Launch Prep

| Deliverable                    | Details                                                                                                      | Acceptance Criteria                                                         |
|--------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Edge server deployment package | Docker Compose stack: orchestrator, PostgreSQL, TimescaleDB, NATS, MinIO, Gazebo. One-command deploy script. | docker compose up starts all services. Health checks pass. < 5 min startup. |
| Cloud deployment (optional)    | Kubernetes manifests for cloud mode. Auto-scaling for analytics and reporting services.                      | Helm chart deploys to K8s. Services scale under load.                       |
| Security audit                 | OWASP top 10 check. JWT validation. Input sanitization. SQL injection prevention. File upload validation.    | Zero critical vulnerabilities. All OWASP checks pass.                       |
| Launch checklist & handoff     | Final QA pass. Known issues documented. Handoff documentation for operations team.                           | All P0 features working. Known issues < 5, none critical. Docs complete.    |

SECTION 11

# Data Model Overview

Core database entities and their relationships. All spatial columns use PostGIS geometry types. Telemetry stored in TimescaleDB hypertables. Media and reports in MinIO object storage.

## Core Entities

| Entity       | Key Fields                                                                                                                 | Storage                      | Notes                                                                                              |
|--------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------|
| Organization | id, name, slug, logo_url, subscription_tier, created_at                                                                    | PostgreSQL                   | Tenant isolation boundary. All data scoped by org_id.                                              |
| User         | id, email, password_hash, role (operator/commander/admin), org_id, last_login                                              | PostgreSQL                   | JWT RS256 auth. Role determines API access and UI visibility.                                      |
| Drone        | id, uin, fc_model, fw_version, total_flight_hours, status, org_id, companion_ip                                            | PostgreSQL                   | Central fleet registry. Status: online/offline/active/maintenance.                                 |
| Payload      | id, drone_id, type (camera/thermal/multispec/sprayer), calibrated_at, status                                               | PostgreSQL                   | Linked to drone. Calibration status affects pre-arm checklist.                                     |
| Mission      | id, name, aoi_geojson (geometry), status, mode (sim/live), org_id, created_by, created_at                                  | PostgreSQL + PostGIS         | + AOI stored as PostGIS polygon. Status: draft/planning/validating/armed/active/completed/aborted. |
| TaskBundle   | id, mission_id, drone_id, waypoints (jsonb), eta_seconds, coverage_pct                                                     | PostgreSQL                   | CBBA output. One bundle per drone per mission.                                                     |
| Geofence     | id, name, type (inclusion/exclusion/ceiling), geom (geometry), alt_ceiling, org_id                                         | PostgreSQL + PostGIS         | + Spatial queries for NFZ conflict checking.                                                       |
| Detection    | id, mission_id, drone_id, model_name, confidence, geom_point (geometry), thumbnail_url, status (pending/accepted/rejected) | PostgreSQL + PostGIS + MinIO | + Thumbnail stored in MinIO. Geom_point + enables spatial heatmap.                                 |
| Telemetry    | timestamp, drone_id, lat, lon, alt, speed, heading, soc, rssi, link_type                                                   | TimescaleDB (hypertable)     | 1 Hz per drone. Partitioned by time. Retention policy configurable.                                |
| AuditLog     | id, actor_id, action, target_type, target_id, details (jsonb), timestamp, ip                                               | PostgreSQL                   | Immutable append-only. All destructive actions logged.                                             |
| SimSession   | id, mission_id, world_sdf_path, speed_multiplier, status, started_at, ended_at                                             | PostgreSQL + MinIO           | + Links to mission. World SDF file stored in MinIO for reproducibility.                            |

|              |                                                                                |                    |                                                                             |
|--------------|--------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------|
| Report       | id, mission_id, type (pdf/json/csv),<br>file_url, generated_at, accessed_count | PostgreSQL + MinIO | File stored in MinIO with signed URL.<br>Access logged to AuditLog.         |
| MissionEvent | id, mission_id, drone_id, event_type, timestamp, details (jsonb)               | PostgreSQL         | Types: detection, battery_swap, link_loss, orca_intervention, replan, note. |

Entity Relationships (Simplified)

- Organization 1—\* User, Organization 1—\* Drone, Organization 1—\* Mission, Organization 1—\* Geofence
- Drone 1—\* Payload, Drone 1—\* Telemetry, Drone 1—\* TaskBundle
- Mission 1—\* TaskBundle, Mission 1—\* Detection, Mission 1—\* MissionEvent, Mission 1—\* SimSession, Mission 1—\* Report
- User 1—\* AuditLog (as actor)

SECTION 12

Go-To-Market Strategy

Target Segments (Priority Order)

| Segment                       | Entry Point                                                  | Expansion Path                                         | Pricing Model                              |
|-------------------------------|--------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------|
| AgriTech Cooperatives         | Crop survey + sprayer automation for 100+ acre farms         | NDVI analytics, seasonal contracts, fleet leasing      | Per-acre per-season SaaS (INR 50–80/acre)  |
| Disaster Response (NDRF/SDRF) | Search & rescue, damage assessment, supply drop coordination | State-level fleet contracts, training programs         | Annual license per fleet (GOI procurement) |
| Infrastructure Inspection     | Solar farm inspection, power line survey, cell tower audit   | Predictive maintenance analytics, compliance reporting | Per-mission or annual enterprise license   |
| Defense R&D; (DRDO/HAL)       | Swarm coordination research platform, simulation environment | Custom perception models, secure deployment            | Custom contract framework (MeITY / DRDO)   |

Go-To-Market Phases

- Phase 1 (Months 1–3): Private beta with 2–3 AgriTech partners. Free Gazebo sim tier. Feedback loop.
- Phase 2 (Months 4–6): Paid AgriTech launch. Pricing: per-acre per-season SaaS model.
- Phase 3 (Months 7–12): Enterprise sales for infrastructure and disaster response. Annual licensing.
- Phase 4 (Year 2): Defense R&D; partnerships. Government procurement framework compliance.

SECTION 13

Competitive Positioning

| Feature                   |           | Drone Matrx         | DJI FlightHub 2 | QGroundControl | DroneUp     | ideaForge SWITCH |
|---------------------------|-----------|---------------------|-----------------|----------------|-------------|------------------|
| Multi-drone (>4)          | swarm     | ✓ (16+)             | ✗               | ✗              | ✓ (limited) | ✓ (limited)      |
| CBBA task allocation      |           | ✓                   | ✗               | ✗              | ✗           | ✗                |
| ORCA avoidance            | collision | ✓                   | ✗               | ✗              | ✗           | ✗                |
| Gazebo pre-flight sim     |           | ✓                   | ✗               | ✗              | ✗           | ✗                |
| Offline-first deploy      | edge      | ✓                   | ✗               | ✓ (single)     | ✗           | Partial          |
| Open FC (PX4 + ArduPilot) |           | ✓                   | ✗ (DJI only)    | ✓              | ✗           | ✗                |
| Real-time video wall      |           | ✓ (multi-feed)      | ✓ (single)      | ✗              | ✓ (single)  | ✓ (single)       |
| ML detection pipeline     |           | ✓ (pluggable ONNX)  | ✗               | ✗              | ✗           | ✗                |
| Mission replay + DVR      |           | ✓ (time-synced)     | Partial         | ✗              | ✗           | Partial          |
| Public API / SDK          |           | ✓ (REST + WS + SDK) | Partial         | ✓ (MAVLink)    | ✗           | ✗                |

KEY DIFFERENTIATORS: Drone Matrx is the only platform combining CBBA swarm allocation, ORCA collision avoidance, and Gazebo SITL pre-flight simulation in a single offline-first product with open flight controller support (PX4 + ArduPilot).

## SECTION 14

## Risks &amp; Mitigations

| Risk                                   |          | Impact | Likelihood | Mitigation                                                                                         |
|----------------------------------------|----------|--------|------------|----------------------------------------------------------------------------------------------------|
| Gazebo resource consumption on edge    |          | High   | Medium     | Profile and optimize. Limit to 8 drones on 32 GB edge. 16 on 64 GB. Auto-throttle sim speed.       |
| PX4 SITL crash during sim              |          | Medium | Medium     | Treat as simulated drone failure. CBBA re-allocates. Log crash dump for debugging.                 |
| MAVLink versioning                     | protocol | Medium | Low        | Pin MAVLink v2. MAVSDK abstracts protocol. FC firmware whitelist enforced.                         |
| CBBA convergence on large swarms (>16) |          | High   | Low        | Algorithm profiled to 16. Phase 2 explores hierarchical CBBA for 32+ drones.                       |
| Connectivity loss in field             |          | High   | High       | Offline-first architecture. Edge server has all mission-critical logic. LoRa heartbeat fallback.   |
| Regulatory compliance (DGCA)           |          | High   | Medium     | Built-in geofence enforcement, NFZ checks, UIN registry, audit logs. Compliance report export.     |
| WebRTC video latency on LTE            |          | Medium | Medium     | Adaptive bitrate. Resolution downgrade on high latency. Coturn TURN server for NAT traversal.      |
| Battery miscalculation                 |          | High   | Low        | Conservative SOC model. RTH triggered at 20% not 10%. Simulated battery validation in Gazebo.      |
| Plugin (ONNX model) stability          |          | Medium | Medium     | Mandatory Gazebo sim validation before live deployment. FPS monitoring. Auto-disable on crash.     |
| Data loss on edge server failure       |          | High   | Low        | PostgreSQL WAL replication. MinIO bucket versioning. NATS JetStream durable streams. Daily backup. |

SECTION 15

Recommended Tech Stack

| Layer              | Technology                                | Justification                                                                             |
|--------------------|-------------------------------------------|-------------------------------------------------------------------------------------------|
| Frontend Framework | React 19 + TypeScript + Vite              | Fast dev server, strong typing, massive ecosystem, team familiarity.                      |
| Styling            | Tailwind CSS 4                            | Utility-first, design token support, dark mode built-in, rapid prototyping.               |
| State Management   | Zustand                                   | Lightweight, no boilerplate, integrates well with React. Sufficient for non-server state. |
| Animation          | Framer Motion                             | Declarative animations, layout transitions, gesture support.                              |
| Map                | Leaflet + React-Leaflet (or Mapbox GL JS) | Open-source, offline tile support, draw plugins, large plugin ecosystem.                  |
| Charts             | Recharts or Chart.js                      | React-native charting. Recharts for declarative API. Chart.js for canvas performance.     |
| Icons              | Lucide React                              | Tree-shakeable, consistent design, MIT licensed.                                          |
| Backend            | Node.js + Fastify                         | High performance, schema-based validation, plugin architecture.                           |
| Auth               | JWT RS256 + bcrypt                        | Stateless auth, refresh token rotation, industry standard.                                |
| Database           | PostgreSQL 16 + PostGIS                   | ACID, spatial queries, JSONB for flexible schemas, mature ecosystem.                      |
| Time-Series        | TimescaleDB                               | PostgreSQL extension. Hypertables for telemetry. Compression. Retention policies.         |
| Object Storage     | MinIO                                     | S3-compatible. Self-hosted. Reports, recordings, thumbnails, ONNX models.                 |
| Message Broker     | NATS JetStream                            | Low latency pub/sub. Durable streams for replay. Lightweight. Offline-capable.            |
| Cache              | Redis                                     | Session cache, rate limiting, real-time leaderboard (if needed), pub/sub bridge.          |
| Drone SDK          | MAVSDK (C++ / Python bindings)            | Official PX4 SDK. Supports MAVLink v2. Async. Multi-drone. Sim and live.                  |
| Simulation         | Gazebo Garden (or Harmonic) + PX4 SITL    | Physics-accurate. Sensor plugins. ROS 2 compatible. Active development.                   |
| Terrain Data       | SRTM (30 m) via GDAL                      | Free global elevation data. GDAL for format conversion to Gazebo heightmap.               |
| ML Inference       | ONNX Runtime (Jetson / Pi)                | Cross-platform. Hardware acceleration on Jetson. CPU fallback on Pi 4.                    |
| Containerization   | Docker + Docker Compose                   | Edge deployment. One-command setup. Reproducible environments.                            |

|               |                |                                                                    |
|---------------|----------------|--------------------------------------------------------------------|
| CI/CD         | GitHub Actions | Free for open-source. Matrix builds. E2E test runners.             |
| E2E Testing   | Playwright     | Cross-browser. Visual regression. Network mocking. CI integration. |
| Documentation | Docusaurus     | MDX support. Versioning. Search. Deployment to GitHub Pages.       |

*END OF DOCUMENT — This PRD is a living document. All sections will be updated as design decisions are finalized, user research validates assumptions, and sprint retrospectives surface new requirements. Next step: use the UI/UX Screen Inventory (Section 08) to create high-fidelity mockups for each screen.*